INCH-POUND

MIL-DTL-32186 (OS)
14 February 2005

DETAIL SPECIFICATION
EXPLOSIVE SEQUENCE CONTROL

This specification is approved for use by the Naval Sea Systems Command. Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers the requirements for the explosive sequence control (ESC), DCU-241/A NAVAIR Drawing 30003-850AS135.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-8500	Interchangeability And Replaceability Of Component Parts For Aerospace Vehicles
MIL-D-21625	Design And Evaluation Of Cartridges For Cartridge Actuated Devices

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A FSC 1377

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NAVAL AIR SYSTEMS COMMAND (CAGE CODE 30003)

850AS135	Explosive Sequence Control DCU-241/A
850AS804	Test Fixture Explosive Sequence Control DCU-241/A
850AS852	Vibration Fixture Explosive Sequence Control DCU-241/A
850AS853	Spacer

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742 Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY FOR AUTOMOTIVE ENGINEERS

SAE-AS5604 Compass, Magnetic, Pilot's Standby

(Copies of SAE documents are available online at http://www.sae.org/servlets/index or from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 USA)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. General requirements

3.1 Preproduction sample. When more than 12 months have elapsed since production of the latest accepted lot, a preproduction sample of 20 explosive sequence controls (ESCs) that have met the requirements of 3.2 through 3.5 shall be required. The sample shall be tested as specified in paragraph 4.2.

3.2 Design and construction. The ESC design and construction shall be identical to that subjected to qualification testing and approved for service use. Selector shall be designed to minimize weight and shall not exceed 1.25 lbs.

3.2.1 Torque Requirements of Threaded Joints. The torque requirements of subassemblies of the device which contain the provisions for the attachment of SMDC or for the mounting or installation of the device

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shall be able to withstand a minimum of 400 inch-pounds of breakaway torque for thread diameters of 1 inch or larger, and a minimum value of one half the maximum torque value specified in Table I for thread sizes under 1 inch in diameter. The torque requirement of other subassemblies of the device which could be loosened or disassembled by handling, vibration, shipment, etc. shall withstand a minimum value of 200 inch-pounds for thread diameters of 1 inch or larger and shall withstand a minimum value of one third of the maximum torque value specified in table I for thread sizes under 1 inch in diameter. See 4.12

TABLE I. Breakaway torque

Bolt or Screw size	Maximum Torque (inch-pounds)
No. 4-40 UNC-3A	3
No. 6-32 UNC-3A	6
No. 8-32 UNC-3A	9
No. 10-32 UNF-3A	13
1/4-28 UNF-3A	30
5/16-24 UNF-3A	60
3/8-24 UNF-3A	80
7/16-20 UNF-3A	100
1/2-20 UNF-3A	150
9/16-18 UNF-3A	200
5/8-18 UNF-3A	300
3/4-16 UNF-3A	400
7/8-14 UNF-3A	600

3.3 Envelope and installation. The dimensions and configuration of the ESC shall be as shown on drawing 850AS135. All parts having the same vendor's item number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500.

3.4 ESC shaft. The ESC shaft shall consist of an HNS Type I transfer booster charge, per 30003-850AS135 note 8, which aligns with the appropriate shielded mild detonating cord (SMDC) outlet port to propagate explosive energy to the selected output SMDC port without detonation of the other output SMDC ports. The explosive loaded shaft subassembly shall be capable of detonating an SMDC test tip across a gap which is four times the maximum production gap. Samples as defined in 4.3.1 shall be tested in accordance with 4.10.

3.5 Inspection

3.5.1 ESC shaft inspection

3.5.1.1 Visual inspection. All ESC shafts manufactured under contract shall be visually inspected and screened for defects as outlined in 4.4.

3.5.1.2 Leakage. ESC shafts shall not exhibit a leak rate in excess of 1×10^{-5} cc per second of air when tested in accordance with 4.5.

3.5.1.3 Radiographic inspection. ESC shafts shall be free of imperfections when examined radiographically in accordance with 4.6.

3.5.2 ESC inspection

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3.5.2.1 Visual inspection. Each ESC manufactured under contract shall be visually inspected and screened for defects as outlined in 4.4.

3.5.2.2 Radiographic examination. ESCs shall be free of imperfections in assembly when examined radiographically in accordance with 4.6.

3.5.2.3 Demagnetization. ESCs shall not deflect an approved pilot's standby compass more than three degrees in either direction when tested in accordance with 4.7.

3.5.2.4 Actuation Force. The total linear force required to position the handle from normal to aft initiates shall be 20 to 30 pounds. The torque required to release the handle from the aft initiates position shall be 10 to 20 inch-pounds. See 4.11.

3.6 Ballistic requirements. The ESCs from each preproduction and production lot shall meet the requirements listed below when subjected to the test prescribed in 4.9.

3.6.1 The ESC when test fired as follows shall initiate the specified outlet SMDC test tip. The remaining two SMDC test tips shall not be initiated, as shown in Table II.

TABLE II. SMDC test tip results.

Handle Position	Inlet Smdc Test Tip Initiated (Port No.)	Outlet Smdc Test Tip Propagated (Port No.)	Outlet Smdc Test Tip Not Propagated (Port No.)
Normal	1	2	3 and 4
Solo	1	4	2 and 3
Aft	1	3	2 and 4

3.6.2 High order detonation. The minimum plate dent for high order detonation of the output SMDC test tip on the witness plate shall be 0.001 inch (see 4.9.1.1).

3.6.3 Misfire. There shall be no misfires. See 4.9.1.2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 20 ESCs shall be prepared using the same procedures and methods proposed for the production lot. Eighteen of these ESCs shall be expended in the tests listed herein and two shall be retained for investigative purposes. A preproduction sample acceptable for functional and environmental testing shall be defined as a sample which has met the requirements of paragraphs 3.2 through 3.5. The contractor shall deliver the ESCs to the activity designated by the contracting agency for tests. The preproduction test sample shall be subjected to the environmental test sequence shown in Table III. The ESCs shall be fired as described in 4.9 and shall meet the functional requirements referenced in 3.6. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction ESCs shall not be applied as part of the quantity specified for delivery by the contract.

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TABLE III. First Article Environmental Test Sequence.

Test Order	Test Description	Plan Para	Test Group Quantity		
			9	9	6
1	Life Cycle	4.13	9	0	0
2	TSHA Handle position normal (in) Handle position Aft (out)	4.14	 4 5	 4 5	 0
3	Visual Inspection	4.14	9	9	
4	Non-Gunfire Vibration Handle position normal (in) Handle position Aft (out)	4.8.1	 4 5	 4 5	 0
5	Gunfire Vibration Handle position normal (in) Handle position Aft (out)	4.8.2	 4 5	 4 5	 0
6	Shock Handle position normal (in) Handle position Aft (out)	4.15	 4 5	 4 5	 0
7	X-Ray Handle position normal (in) Handle position Aft (out)	4.6	 4 5	 4 5	 0
8	N-Ray Handle position normal (in) Handle position Aft (out)	4.6	 4 5	 4 5	 0
Functional Testing					
9	High Temp (+200ºF) Handle position normal (in) Handle position Aft (out) Handle position Solo (spacer)	4.9	 1 1 1	 1 1 1	 1 1
10	Low Temp (-65ºF) Handle position normal (in) Handle position Aft (out) Handle position Solo (spacer)	4.9	 1 1 1	 1 1 1	 1 1
11	Ambient Temp (+65º to +95ºF) Handle position normal (in) Handle position Aft (out) Handle position Solo (spacer)	4.9	 1 1 1	 1 1 1	 1 1

4.3 Conformance inspection. All sequencers shall have met the requirements of paragraphs 3.2 through 3.5 as a prerequisite to conformance inspection. The conformance inspection shall consist of the tests in 4.4 through 4.7, and 4.9 through 4.12.

4.3.1 Sampling. Samples for ballistic test shall be selected from the production lot in accordance with the requirements of Table I of general inspection level II of ANSI/ASQ Z1.4. Two additional ESCs shall be provided with each test sample for investigative purposes. Sample ESCs and ESCs made available for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or order. The production lot size shall be 26 minimum and 280 maximum. Samples for shaft gap testing shall be 2 each for lots of 26 to 50, 4 each for lots of 51 to 90 and 6 each for lots larger than 90. Shaft gap test shall be conducted in accordance with 4.10.

4.4 ESC and esc shaft visual inspection. Visually examine all ESC shafts prior to ESC assembly and visually examine all assembled ESCs. Reject those with inadequate or illegible descriptive marking or serial numbers, dents, deep scratches, or other defects which may prevent proper installation or function.

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4.5 ESC shaft leak test. Each ESC shaft produced under contract shall be leak tested in a dry gas leak tester to determine conformance to 3.5.1.2. Units which exhibit a leak rate in excess of $1x10^{-5}$ std cc/sec of air at a pressure differential of 1.0 ± 0.1 atmosphere shall be rejected.

4.6 Radiographic examination

4.6.1 ESC shaft examination. All ESC shafts shall be identified with serial numbers prior to radiographic examination. The ESC shafts shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the ESC shafts displayed thereon. The ESC shafts shall be positioned to give the most revealing cross-section of the shaft. The radiograph identification shall include the complete lot number for the ESC shafts and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective shafts found by the vendor shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test (see 5.1).

4.6.1.1 X-Ray examination. Each ESC shaft shall be x-rayed prior to ESC assembly, in accordance with ASTM E1742, to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for rejection.

4.6.1.2 N-Ray examination. Each ESC shaft shall be N-rayed, prior to ESC assembly, to confirm the presence of the explosive, non-metal parts, no foreign hydrogenous materials and the proper assembly of pyrotechnic/non-metallic materials. Any observable imperfection shall be cause for rejection.

4.6.2 ESC examination. All assembled ESCs shall be identified w1th serial numbers prior to radiographic examination. The ESCs shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the ESCs displayed thereon. The ESCs shall be positioned to give the most revealing cross-section of the ESC. The radiograph identification shall include the complete lot number, as stamped on the ESCs, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective ESCs found by the vendor shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test (see 5.1).

4.6.2.1 X-Ray examination. Each assembled ESC shall be x-rayed in accordance with ASTM E1742, to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for rejection.

4.7 Demagnetization. Each ESC shall be moved past a pilot's standby compass face (per SAE AS5604) by placing the ESC six inches from the compass face, in a north-south horizontal position with the compass set in an east-west heading. The ESC and the compass shall be placed at the same height from the floor. The ESC shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved back past the compass face. This procedure shall be repeated for each 90 degrees rotation of the ESC.

4.8 Vibration test

4.8.1 Non-Gunfire Random & Sinusoidal Requirements. The combined random and sinusoidal (non-gunfire) requirements involve a combination of sinusoidal cycling (5 to 50 Hz) and high frequency (100 to 2000 Hz) random vibration. The procedures shall be conducted along all three orthogonal axes at the following temperatures; -65ºF, +70ºF, and +200ºF. The non-gunfire random and sinusoidal vibration tests

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shall be performed on each of the ESCs. Two accelerometers shall be used to monitor the response of the ESCs during the test. One accelerometer shall be located on a handle and the other on a body of one of the ESCs along the centerline of port no. 4 The handle position for the testing shall be as specified in Table III.

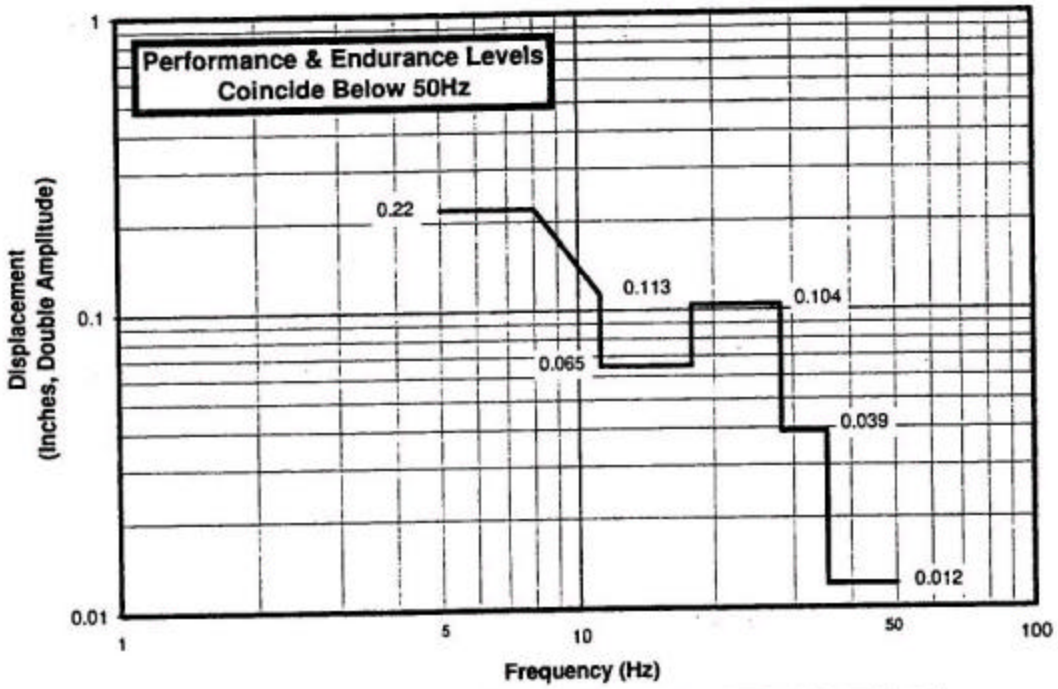
4.8.1.1 Sinusoidal Vibration Test. The ESCs shall be vibrated at 70ºF along the same orthogonal axis between 5 Hz to 50 Hz and returning to 5 HZ in approximately 7.5 minutes for a total time of 20 minutes per axis per figure 1. Repeat at -65ºF and again at +200ºF.

4.8.1.2 Random Vibration Test. The ESCs shall be vibrated along each orthogonal axis at 70°F for 40 minutes per axis for a total of 120 minutes in accordance with the applicable random vibration spectrum shape, endurance test levels per figure 2 and Table IV. Repeat at -65°F, and then again at +200°F.

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VIBRATION (NON-GUNFIRE) - RANDOM/SINUSOIDAL LOW FREQUENCY SINUSOIDAL TEST LEVELS (5 - 50 Hz) REGION V-FF-3

(Cockpit)



Non-Gunfire Vibration Region V-FF-3 Sinusoidal	
Frequency Range (Hz)	Displacement or Acceleration
5-8	0.22 in D.A.
8-11.2	±0.7 G
11.2-17.8	0.065 in D.A.
17.8-28.2	0.104 in D.A.
28.2-35.5	0.039 in D.A.
35.5-50	0.012 in D.A.

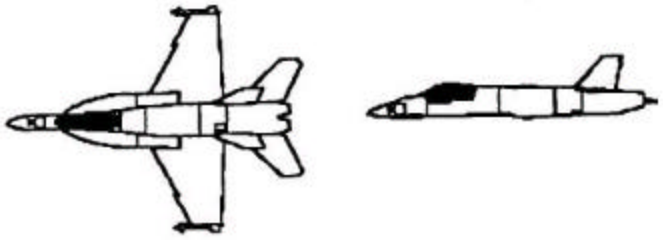
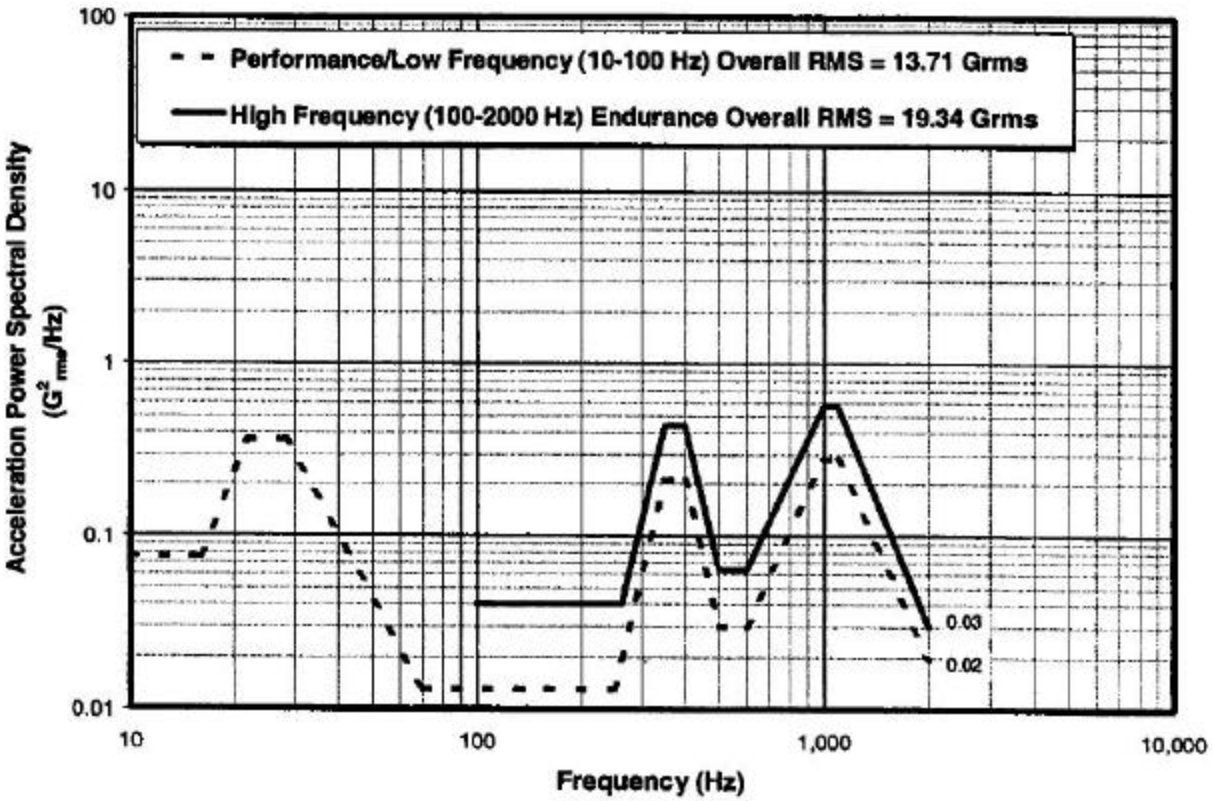


FIGURE 1. Low Frequency Vibration Profile

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Note: Use High Frequency Endurance levels

FIGURE 2. Non-Gunfire Random Vibration Test Profile

TABLE IV. Non-Gunfire Random Vibration Test levels

F/A-18 E/F Non-gunfire Zone V-FF-3 Random			
Performance: Overall 13.71 G_{rms}		Endurance: Overall 19.34 G_{rms}	
Frequency Range (Hz)	G^2_{rms}/Hz or dB/Octave	Frequency Range (Hz)	G^2_{rms}/Hz or dB/Octave
10-16	0.076 G^2_{rms}/Hz	100-262	0.040 G^2_{rms}/Hz
16-22	13.992 dB/Octave	262-350	24.944 dB/Octave
22-28	0.363 G^2_{rms}/Hz	350-400	0.435 G^2_{rms}/Hz
28-70	-11.027 dB/Octave	400-500	-26.144 dB/Octave
70-250	0.013 G^2_{rms}/Hz	500-600	0.063 G^2_{rms}/Hz
250-350	24.944 dB/Octave	600-1000	13.056 dB/Octave
350-400	0.211 G^2_{rms}/Hz	1000-1100	0.575 G^2_{rms}/Hz
400-500	-26.144 dB/Octave	1100-2000	-14.646 dB/Octave
500-600	0.030 G^2_{rms}/Hz		
600-1000	13.056 dB/Octave		
1000-1100	0.279 G^2_{rms}/Hz		
1100-2000	-14.646 dB/Octave		

Note: Use Endurance levels

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4.8.2 Gunfire Sinusoidal Vibration. A sinusoidal vibration test at +70ºF is used in the frequency range of 50 to 2000 Hz. Testing shall consist of logarithmic sweeps from 50 to 2000 Hz and back to 50 Hz in 15 minutes plus fixed dwell periods per figures 3 and 4. The gunfire vibration tests shall be performed on each of the ESCs. Two accelerometers shall be used to monitor the response of the ESCs during the test. One accelerometer shall be located on a handle and the other on the body of one of the ESCs along the centerline of port no. 4. The handle position for the testing shall be as specified in Table III.

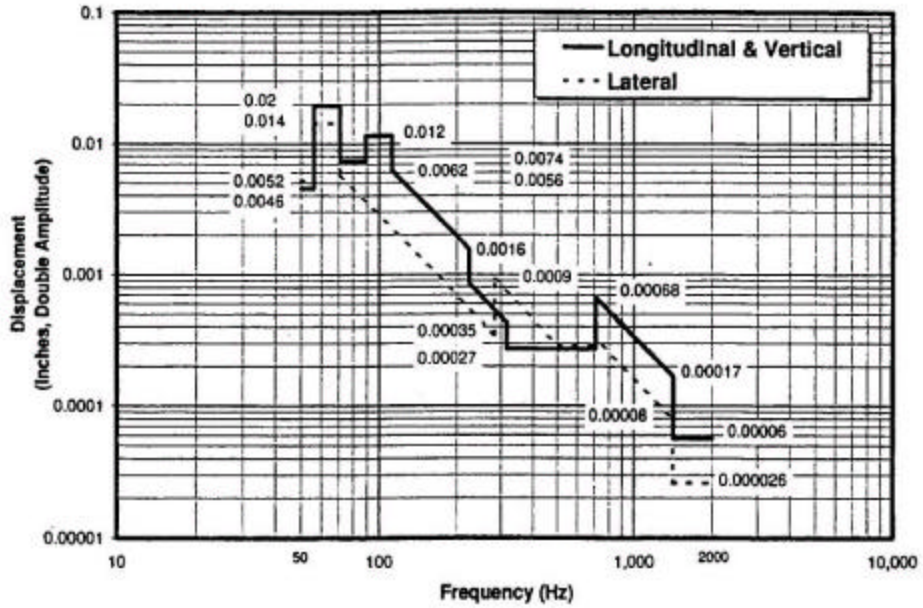
4.8.2.1 Vibration Cycling. The items shall be vibrated along the same orthogonal axis, with the frequency varying at a logarithmic rate from 50 to 2000 Hz and back to 50 Hz in 15 minutes. Double amplitude or vibratory acceleration levels shall be in accordance with figure 3.

4.8.2.2 Fixed Dwell. The initiators shall be vibrated along the same orthogonal axis at the fixed narrow band dwell frequencies. The fixed narrow band dwells shall be performed by sweeping the frequencies about ±5% of the specified center frequency, see figure 4. The sweep rate shall be approximately the same as in the above paragraph. The vibration test levels shall be as shown in figure 4.

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VIBRATION (GUNFIRE) - SINUSOIDAL TEST LEVELS
REGION V-GF-5

(Forward Cockpit - Main Instrument Panel and Vertical Consoles)



Gunfire Vibration Region V-GF-5 Sinusoidal			
Longitudinal & Vertical		**Lateral**	
Frequency Range (Hz)	Displacement or Acceleration	Frequency Range (Hz)	Displacement or Acceleration
50-56 Hz	0.0046 in D.A.		
56-71 Hz	0.020 in D.A.	50-56 Hz	0.0052 in D.A.
71-89 Hz	0.0074 in D.A.	56-71 Hz	0.014 in D.A.
89-112 Hz	0.012 in D.A.	71-283 Hz	±1.4 G
112-224 Hz	±4.0 G	283-505 Hz	±3.7 G
224-315 Hz	±2.2 G	505-750 Hz	0.00029 in D.A.
315-707 Hz	0.00027 in D.A.	750-1414-Hz	±8.2 G
707-1414 Hz	±17.3 G	1414-2000 Hz	0.000026 in D.A.
1414-2000 Hz	0.000057 in D.A.		

FIGURE 3. Vibration (Gunfire) - Sinusoidal Test Schedule

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Sweep time (min/axis)	60.0
Fixed Dwell (f_d) at the following Center frequencies (see note 1)	
67 Hz (63.7-70.4 Hz)	7.5
100 Hz (95-105 Hz)	7.5
135 Hz (128-142 Hz)	7.5
200 Hz (190-210 Hz)	15.0
267 Hz (254-280 Hz)	7.5
300 Hz (285-315 Hz)	7.5
335 Hz (318-352 Hz)	7.5
400 Hz (380-420 Hz)	15.0
467 Hz (444-490 Hz)	7.5
500 Hz (475-525 Hz)	7.5
Total Time Each Axis (minutes)	150.0

FIGURE 4. Vibration (Gunfire) - Sinusoidal Test Schedule

4.9 Ballistic test. The ESCs shall be selected from each production lot in accordance with paragraph 4.3.1. The sample of ESCs shall be divided, as nearly as possible, into three equal groups. One group shall be fired, with the handle in the normal position, the other group shall be fired with the handle in the solo position and the remaining group shall be fired with the handle in the aft position. If the sample of ESCs is not equally divisible by three, the excess of ESCs shall be placed in the group to be fired with the handle in the normal position. The group to be fired with the handle in the solo position shall have a solo spacer (NAVAIR DWG NO. 850AS853) installed. Each group shall be subdivided, as nearly as possible, into three equal subgroups. If the sample of ESCs is not equally divisible by three, the excess of ESCs shall be placed in the subgroup to be fired at –65°F. One subgroup shall be fired at -65°F, the other shall be fired at +70°F and the remaining subgroup shall be fired at +200°F. One subgroup each shall be conditioned, with the test tips installed, for a period of time neither less than 4 hours nor greater than 24 hours at -65°F ± 5°F and +200°F ± 5°F. The remaining subgroup shall be conditioned, with the test tips installed, for a period of time not less than 4 hours at +70°F ± 5°F. After temperature conditioning, the handle shall move freely between the normal, solo and aft positions. The ESC shall be fired in the test fixture NAVAIR DWG NO. 850AS804, as shown on drawing 850AS135 within five minutes after removal from the temperature conditioning chamber. If any ESC is exposed to the temperature environment for greater than the time allowed or not fired within five minutes after removal, it shall be returned to a +70°F ± 5°F environment for at least 4 hours and then reexposed to the original environment for not less than 4 hours and no more than 24 hours before test firing. The reexposure may be performed as many times as necessary to complete the required ballistic test. Instrumentation shall be provided to measure and record the high order detonation of the designated ESC output SMDC test tip. The production lot shall be accepted if the sample fulfills the following criteria.

4.9.1 Acceptance criteria.

4.9.1.1 High order detonation. The high order detonation of the output SMDC test tip propagated, as specified in paragraph 3.6.2 herein, shall be demonstrated in a 6061-T6 aluminum witness plate. The lot shall be acceptable with respect to high order detonation if the ballistic sample meets the requirements of 3.6.2. The acceptable number of defects is 0 and the rejection number is 1 or more.

4.9.1.2 Misfire. The ESC that does not fire under the conditions specified in 4.9 shall be considered defective and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

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4.9.2 Retest. There shall be no retests. See 4.9.3.

4.9.3 Test failure. If test failure is attributable to an assignable cause, excluding the test ESC, the original test results shall be discarded and that test reconducted.

4.10 Shaft Gap Test. Prior to assembly of the selector, shafts shall be selected at random as defined in 4.3.1 and subjected to ambient temperature testing at a distance four times the maximum production gap.

4.11 Actuation Force. Each sequencer shall be cycled from the normal position to the aft initiates position, locked in place and then returned to the normal position. The actuation force and torque shall be in accordance with 3.5.2.4.

4.12 Breakaway Torque. Each preproduction test sample and lot acceptance test sample shall pass the breakaway torque test values specified in 3.2.1. The break away torque test shall be conducted by the contractor unless otherwise specified in the contract.

4.13 Life Cycle. First article test specimens specified in Table III shall be exposed to physical life cycle testing of the Selector. The Selector handle shall be cycled 3000 times, each cycling representing the full drawn out position of the handle, then a 90º turn of the handle, the rotation back 90ºs and the release of the handle to its original position. The items shall not fire or incur damage or degradation during or subsequent to the life cycle test.

4.14 Temperature Shock Humidity and Altitude Cycling. First article test specimens specified in Table III shall be exposed to a four (4) week (28-day) temperature shock, humidity and altitude cycling environment as described below, in accordance with the applicable portion of MIL-D-21625

The test items with stainless steel plugs and seals installed in the inlet and outlet ports shall be submitted to the TSH&A cycling test outlined in Table III and the following requirements:

1. The test items shall be positioned in the test chamber vertically with the handle upward.
2. The handle position for the testing shall be as specified in Table III.
3. No test item shall be in contact with another test item during the cycling shown in table V.
4. There shall be no interim-withdraws of units.

Table V. Temperature Shock/Humidity and Altitude Cycling Schedule

Monday	0800	Place sequencers in chamber at 70°F at 50% relative humidity (RH).
	1200	Raise chamber temperature to 160°F and an RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi)
Tuesday	0800	Remove the sequencers from above chamber and immediately place in chamber maintained at 70°F at 50% RH.
	1200	Raise chamber temperature to 160°F and an RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove sequencers from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.

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Table V. Temperature Shock/Humidity and Altitude Cycling Schedule (cont'd)

Wednesday	0800	Reduce chamber temperature to 70°F and RH to 50%. The chamber temperature shall reach 70°F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160°F and an RH of 95%. The chamber temperature shall reach 160°F no later than 1300.
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi).
Thursday	0800	Remove the sequencers from above chamber and immediately place in a chamber maintained at 70°F at 50% RH.
	1200	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Friday	0800	Reduce chamber temperature to 70°F and RH to 50%. The chamber temperature shall reach 70°F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160°F and RH to 95%. The chamber temperature shall reach 160°F no later than 1300.
	1600	Remove the sequencers from above chamber and immediately place in a chamber maintained at -65°F at a standard ambient pressure.

This schedule shall be followed for a total of four (4 weeks) (28 days) except that on the second and fourth weekend, the soak time shall be from 1200 on Friday until 0800 Monday at a temperature of 160ºF at 95% RH.

Success criteria: The items shall not incur detrimental degradation or damage during the TSHA test and shall not allow moisture intrusion. Evidence of corrosion on internal components is not acceptable. When TSH&A testing is complete, items shall be subjected to visual inspection prior to vibration testing. Port plugs shall be removed for this inspection.

4.15 Shock. First Article Test specimens specified in Table III shall be exposed to a service shock environment in accordance with MIL-DTL-21625. The handle position for the testing shall be as specified in Table III.

The shock pulse waveform shall be a terminal peak saw tooth. the peak amplitude shall be 20 g's and the duration shall be 11 milliseconds. Three shocks shall be applied in each direction of the three mutually perpendicular axes, for a total of eighteen shocks.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The explosive sequence control (ESC), DCU-241/A was designed to select the ejection mode (normal, solo or aft) for the TC-18 aircraft aircrew escape system. Since the ESC was designed strictly for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

A. Title, number and date of source control drawing 850AS135, this specification and SPI 14083-520-174-0030.
B. Whether preproduction inspection is required, and if so, specify the test activity.
C. Production lot size and test activity.
D. Direction for shipping radiographic plates of entire production lot along with ballistic sample (paragraph 3.6) to the activity conducting production lot acceptance test.
E. Items of data required for each preproduction and production lot.

6.2.1 Safety precautions. The safety precautions requirements of the "Contractor's Safety Manual For Ammunition, Explosives And Related Dangerous Material" (DOD 4145.26m) are applicable and should be specified in the contract as required by the Defense Acquisitions Regulation (DAR) 1-323. When this specification is used as part of The description of work to be accomplished by a government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.3 Explanation of terms. For the purpose of this document, NAVSUPINST 4030.28a (20 June 1975) paragraph 1-4, 2-2,2-3, and Appendix A, when applicable should be used to clarify terms.

6.4 Subject term (keyword) listing.

TC-18 aircraft
Aircrew escape system.

Preparing activity
Navy – OS
(Project no. 1377-0236)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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